ENGINE OIL SOLENOID VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

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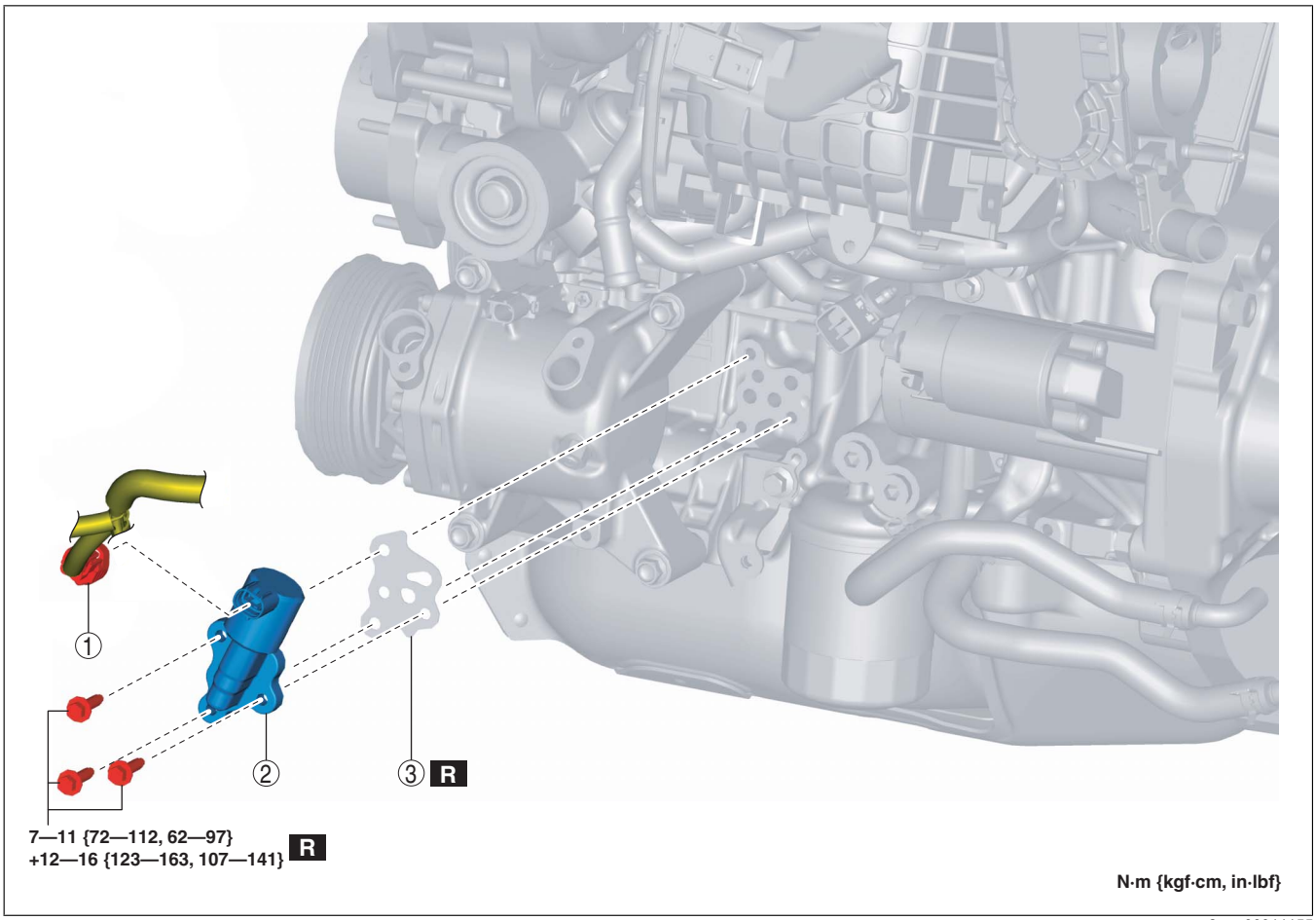
Warning

- **Hot engines and engine oil can cause severe burns. Turn off the engine and wait until it and the engine oil have cooled.**
- **A vehicle that is lifted but not securely supported on safety stands is dangerous. It can slip or fall, causing death or serious injury. Never work around or under a lifted vehicle if it is not securely supported on safety stands.**
- **Continuous exposure to USED engine oil has caused skin cancer in laboratory mice. Protect your skin by washing with soap and water immediately after working with engine oil.**

Note

- Remove the electric water pump before removing/installing the engine oil solenoid valve.

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the front under cover No.1 and No.2. (See FRONT UNDER COVER No.1 REMOVAL/ INSTALLATION.)(See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
4. Drain the following coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)(See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
   - — ATX: Engine coolant and water-cooled charge air cooler coolant
   - — MTX: Water-cooled charge air cooler coolant
5. Remove the electric water pump. (See ELECTRIC WATER PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Remove in the order indicated in the table.
7. Install in the reverse order of removal.
8. Refill the following coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)(See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
   - — ATX: Engine coolant and water-cooled charge air cooler coolant
   - — MTX: Water-cooled charge air cooler coolant
9. Inspect the following coolant leakage.(See ENGINE COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].) (See WATER-COOLED CHARGE AIR COOLER COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].)
   - — ATX: Engine coolant and water-cooled charge air cooler coolant
   - — MTX: Water-cooled charge air cooler coolant
10. Start the engine and confirm that there is no oil leakage.
    - If there is oil leakage, repair or replace the applicable part.



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1	Engine oil solenoid valve connector
2	Engine oil solenoid valve (See Engine Oil Solenoid Valve Removal Note.) (See Engine Oil Solenoid Valve Installation Note.)
3	Gasket

Engine Oil Solenoid Valve Removal Note

1. Cover the area around the engine oil solenoid valve with a rag because a small amount of engine oil will leak from the engine oil solenoid valve installation area.

Engine Oil Solenoid Valve Installation Note

Note

- The three bolts can be tightened in any order.

1. Tighten the three new bolts in two steps.

Tightening torque

Step 1: 7—11 N·m {72—112 kgf·cm, 62—97 in·lbf}

Step 2: 12—16 N·m {123—163 kgf·cm, 107—141 in·lbf}